

CHIBA, Japan

WMO: 476820

Lat: 35.6025N

Lon: 140.1044E

Elev: 20

StdP: 14.69

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.8	34.3	7.2	8.0	41.0	9.9	9.1	41.5	30.2	50.6	27.1	48.4	7.5	340	0.469

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.8	90.7	78.4	88.9	77.9	87.2	77.3	80.0	87.4	79.1	86.2	78.4	85.1	12.2	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.0	145.4	84.5	77.1	141.4	83.8	76.4	137.6	83.3	43.6	87.6	42.7	86.4	42.0	85.3	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.8	22.8	19.5	DB	29.9	95.5	1.3	2.4	29.0	97.2	28.2	98.6	27.5	99.9	26.6	101.7	(4)
(5)			WB	24.7	81.6	1.6	1.0	23.6	82.3	22.7	82.9	21.8	83.4	20.7	84.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.7	43.0	43.8	49.4	58.3	66.5	71.9	79.2	81.8	75.6	66.2	56.2	47.5	(6)	
	DBStd	14.14	3.78	4.72	5.73	5.80	4.78	4.47	4.96	4.27	5.44	4.97	5.07	4.59	(7)	
	HDD50	606	220	183	83	7	0	0	0	0	0	0	7	106	(8)	
	HDD65	2865	682	593	484	209	40	4	0	0	1	44	266	542	(9)	
	CDD50	4884	3	10	64	257	512	657	906	985	767	501	193	29	(10)	
	CDD65	1667	0	0	0	8	87	211	441	520	318	80	2	0	(11)	
	CDH74	13306	0	0	0	9	205	893	4109	5537	2320	232	1	0	(12)	
	CDH80	4166	0	0	0	0	12	123	1333	2086	591	21	0	0	(13)	
(14) Wind	WSAvg	8.8	8.5	8.8	9.5	10.1	9.7	8.7	9.5	9.4	8.6	7.9	7.3	7.9	(14)	
(15) Precipitation	PrecAvg	57.8	2.6	2.6	4.3	4.5	5.0	5.9	5.3	5.0	8.1	8.0	4.2	2.3	(15)	
	PrecMax	76.9	6.1	5.3	7.6	8.2	9.2	12.0	12.4	12.5	16.7	23.8	10.3	7.9	(16)	
	PrecMin	40.3	0.1	0.2	1.3	1.7	2.3	1.7	0.6	0.4	1.7	0.9	0.4	0.1	(17)	
	PrecStd	8.8	1.7	1.3	1.5	1.8	1.8	2.2	3.5	3.3	4.1	5.4	2.4	1.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.8	64.8	68.4	75.1	81.8	86.7	93.0	94.5	89.8	82.7	72.4	66.8	(19)	
		MCWB	52.2	57.1	57.0	61.9	65.9	72.6	77.8	78.9	77.0	70.1	63.6	59.9	(20)	
	2%	DB	55.7	59.2	65.2	71.9	78.2	83.3	90.1	91.4	87.4	79.0	69.8	61.6	(21)	
		MCWB	47.4	52.1	56.6	61.5	65.4	72.2	78.0	79.1	76.7	69.9	61.2	52.9	(22)	
	5%	DB	52.9	55.4	62.3	69.5	76.2	80.8	88.3	89.8	85.5	76.5	67.4	58.5	(23)	
		MCWB	43.8	47.4	54.1	60.1	65.0	71.2	77.6	78.5	76.1	67.9	59.2	49.9	(24)	
	10%	DB	50.6	52.3	59.6	67.5	74.2	78.9	86.4	88.1	83.5	74.1	65.0	55.8	(25)	
		MCWB	41.5	44.0	51.8	58.7	64.3	70.6	77.1	78.1	75.2	66.0	57.1	47.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.6	58.8	61.2	67.8	72.3	76.7	80.8	81.7	79.6	74.7	67.1	61.5	(27)	
		MCDB	58.3	64.4	64.7	71.4	76.3	83.9	88.8	90.4	85.8	79.1	70.8	66.0	(28)	
	2%	WB	48.8	53.2	58.4	64.3	69.5	75.1	79.5	80.4	78.5	72.2	63.5	55.0	(29)	
		MCDB	53.9	58.0	63.3	69.6	74.3	80.2	87.0	88.0	84.5	77.0	67.7	60.5	(30)	
	5%	WB	45.7	48.7	55.8	62.1	67.8	73.8	78.7	79.6	77.6	69.9	61.1	51.2	(31)	
		MCDB	51.4	54.3	61.4	67.5	73.2	78.9	85.8	86.9	83.3	74.6	65.4	56.5	(32)	
	10%	WB	43.0	45.2	52.8	60.1	66.2	72.2	77.8	79.0	76.6	67.6	58.8	48.7	(33)	
		MCDB	48.9	50.8	58.5	65.5	71.8	77.1	84.5	85.9	82.2	72.6	63.3	54.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	13.0	13.4	13.2	12.3	10.7	10.4	10.8	10.5	11.1	12.3	12.7	(35)	
		MCDBR	15.8	16.9	15.9	15.0	14.7	13.0	12.4	12.1	11.6	13.0	14.8	15.8	(36)	
	5% WB	MCWBR	11.7	13.3	12.1	9.2	7.6	6.1	5.1	4.6	5.1	7.9	10.8	12.2	(37)	
		MCWBR	15.1	16.4	15.1	13.1	12.3	10.7	11.1	11.4	10.7	11.8	13.4	14.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.384	0.434	0.503	0.551	0.569	0.596	0.589	0.573	0.510	0.484	0.431	0.387	(40)	
		taud	2.233	2.086	1.926	1.853	1.868	1.868	1.932	1.964	2.106	2.105	2.166	2.243	(41)	
	Ebn at Noon	Ebn at Noon	251	250	242	238	236	229	229	230	239	231	231	241	(42)	
		Edh at Noon	34	44	57	64	64	64	60	57	47	43	36	32	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	865	1013	1276	1493	1610	1428	1576	1594	1223	948	780	756	(44)	
	RadStd	RadStd	62	107	83	149	168	133	266	148	107	96	73	60	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+124	N/A
Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page